

CRUSTACEANS

Most crustaceans are aquatic, breathe with gills, and have limbs for crawling or swimming. A few are sedentary as adults, parasitic, or land living.

The basic body plan of a crustacean consists of a head, thorax, and abdomen, though in many groups the head and thorax are fused together. In crabs, lobsters, and prawns a frontal flap grows backwards to cover the entire head and thorax with a shell-like carapace. Crustaceans are the only arthropods that have two pairs of antennae and primitively two-pronged limbs. Limbs on the thorax are usually used for locomotion, but in some species they are modified as pincers for feeding or defence. Many crustaceans also have well-developed abdominal limbs, which are

often used for brooding young. Gills tie most crustaceans to water, although woodlice and some crabs have modified breathing organs that enable them to live in damp places on land.

Being permanently under water, where their weight is supported, means that crustaceans can develop a thick, heavy exoskeleton. Many species have an exoskeleton hardened with minerals, which are reabsorbed between moults when the outer coating of the body is shed. With their body buoyed up by water, many crustaceans have become bigger than their land-based relatives. The world's largest arthropod is the deep-sea Japanese spider crab (leg span 4m/13ft). At the other extreme, crustaceans make up the bulk of animal life in plankton, either as tiny larvae or as adult shrimps and krill.

PHYLUM	ARTHROPODA
SUBPHYLUM	CRUSTACEA
ORDERS	56
FAMILIES	About 1,000
SPECIES	About 70,000

DEBATE CRUSTACEAN ANCESTRY

Crustaceans have many unique features, including double antennae and two-pronged limbs, that suggest they are exclusive descendants of a single common ancestor. However, recent DNA analysis shows that insects are also descended from the crustacean group.

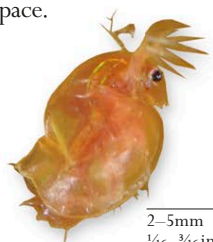
WATER FLEAS AND RELATIVES

The class Branchiopoda are primarily freshwater crustaceans, which make up part of the plankton that flourish in short-lived pools, surviving long periods of drought as eggs. Their thorax carries leafy limbs for breathing and filter feeding. Water fleas are typically encased in a transparent carapace.



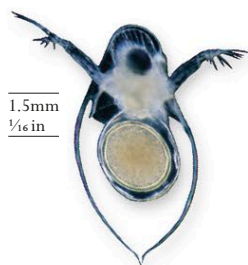
BRINE SHRIMP
Artemia salina
F: Artemiidae

This soft-bodied, stalk-eyed animal is found worldwide in salt pools and swims upside down. Its hard-shelled eggs can resist years of drought.



LARGE WATER FLEA
Daphnia magna
F: Daphniidae

This N. American water flea, like its relatives, can brood eggs inside its carapace. These hatch without fertilization, rapidly populating ponds as a result.



MARINE WATER FLEA
Evdne nordmanni
F: Podonidae

Most water fleas – named for their jerky swimming – occur in stagnant freshwater pools. This, however, is a saltwater species of oceanic plankton.

VERNAL POOL TADPOLE SHRIMP
Lepidurus packardii
F: Triopsidae

Tadpole shrimps are ancient, bottom-living crustaceans of temporary freshwater pools. Relatives of this Californian species have changed little in 220 million years of evolution.



two tails

BARNACLES AND COPEPODS

Like other marine crustaceans, the class Maxillopoda start life as tiny planktonic larvae. Barnacle larvae cement themselves, head down, on rocks. Most copepods stay free-swimming, although some become parasites.

0.5–1.5 cm
3/16–1/2 in



COMMON ACORN BARNACLE
Semibalanus balanoides
F: Archaeobalanidae

Sensitive to drying out, this intertidal species is most abundant at the base of the barnacle zone on exposed North Atlantic rocky coastlines.

1.8 cm
3/4 in



SEA LOUSE
Caligus sp.
F: Caligidae

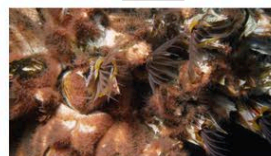
A representative of a group of copepods that are parasitic on marine fish, this crustacean attacks salmon and related species.

3–5 mm
1/8–3/16 in

GLACIAL COPEPOD
Calanus glacialis
F: Calanidae

Within Arctic Ocean plankton, this copepod is an important part of the food chain.

5–10 cm
2–4 in



GIANT ACORN BARNACLE
Balanus nubilus
F: Balanidae

The world's largest barnacle, this species occurs attached to rocks below the intertidal level along the Pacific N. American coast.



ASIAN ACORN BARNACLE
Tetraclita squamosa
F: Tetraclitidae

This acorn barnacle lives in the intertidal zone on Indo-Pacific shores. Recent research has led to five subspecies being recognised.

DEEP SEA GOOSE BARNACLE
Neolepas sp.
F: Eolepadidae

This goose barnacle ally lives around volcanic vents on the ocean floor, where it filter-feeds on other organisms including bacteria.

5–10 cm/2–4 in



COMMON GOOSE BARNACLE
Lepas anatifera
F: Lepididae

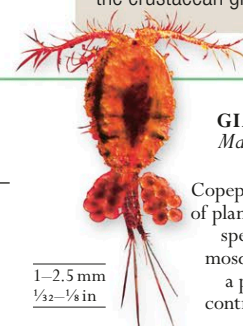
Goose barnacles are attached – mostly to oceanic flotsam – by a flexible stalk. This species occurs in temperate waters of the N.E. Atlantic.

COMMON FISH LOUSE
Argulus sp.
F: Argulidae

This is a flattened, fast-swimming crustacean with an oval carapace. It uses suckers to attach to fish and suck their blood.



0.5–1 cm
1/16–3/8 in



GIANT COPEPOD
Macrocylops albidus
F: Cyclopidae

Copepods are tiny predators of plankton; this widespread species even preys on mosquito larvae, offering a potential means of controlling these insects.

